Why scan enable

Scan enable muxes functional vs scan data; SI/SO bookend the chain

The idea

- GOLDEN: se=scan_enable, si=scan_in, so=scan_out

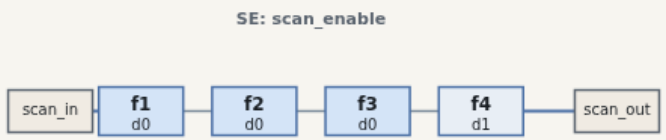
SE port

SCAN ENABLE

SE port

Step 1 / 5 · se · module01-03-scan-enable

Scan chain (left → right)



SE port

Explanation

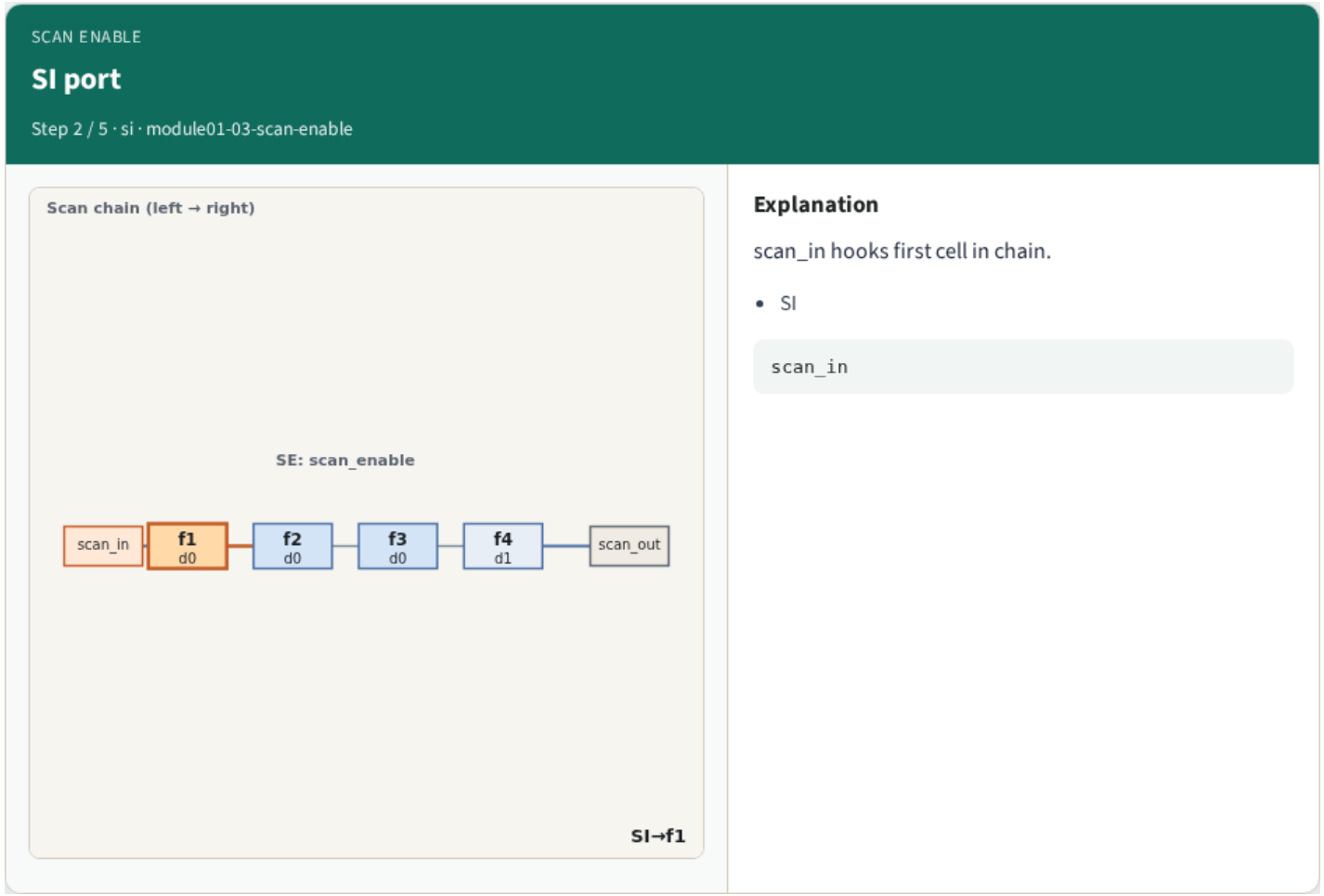
scan_enable drives mux select on all SDFFs.

- SE

```
scan_enable
```

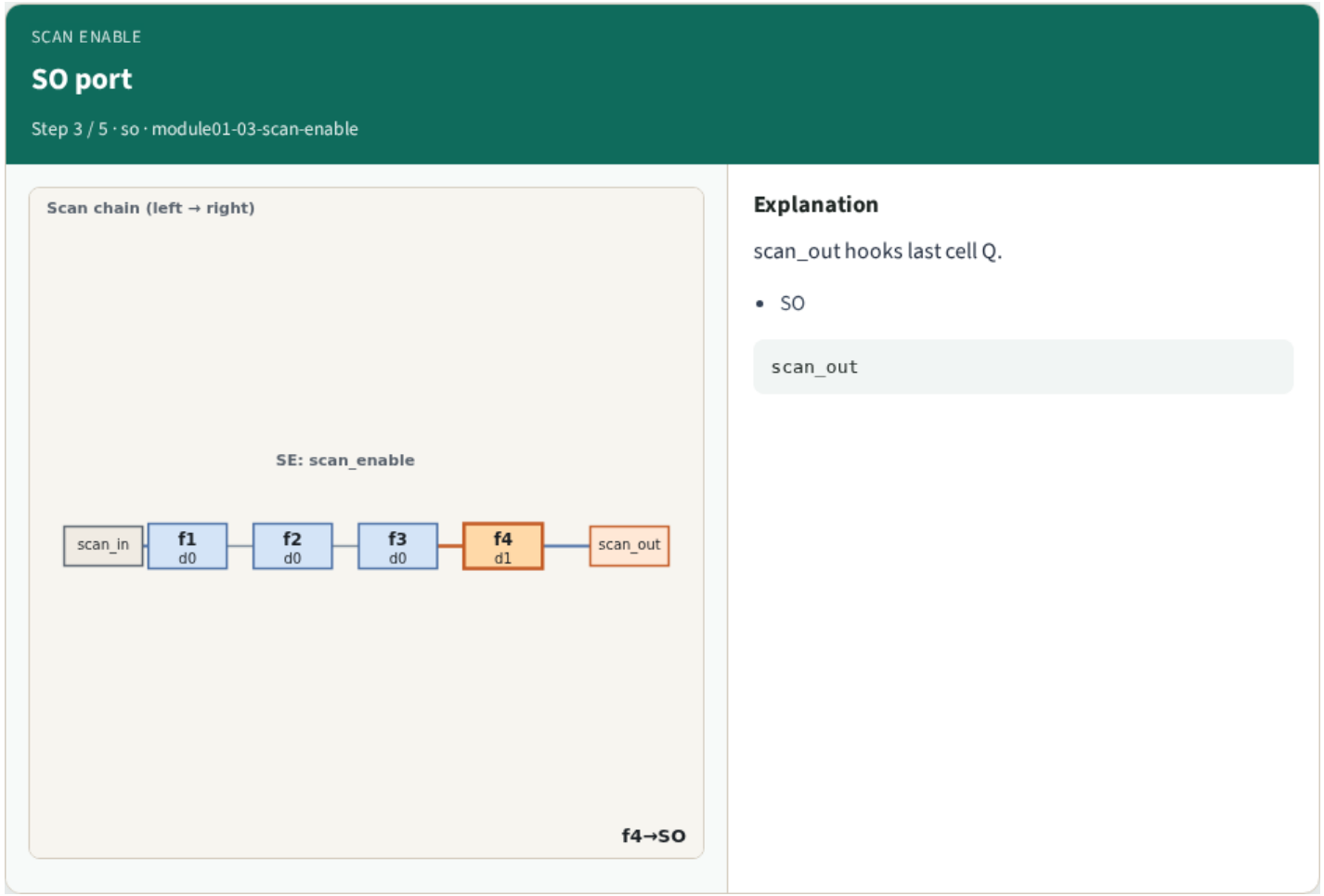
SE port

SI port



SI port

SO port



SO port

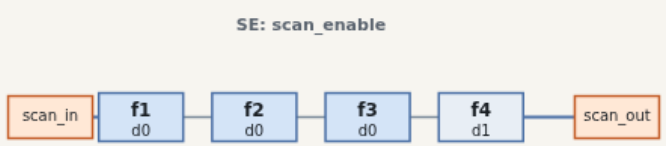
Port names

SCAN ENABLE

Port names

Step 4 / 5 · hook · module01-03-scan-enable

Scan chain (left → right)



3 ports

Explanation

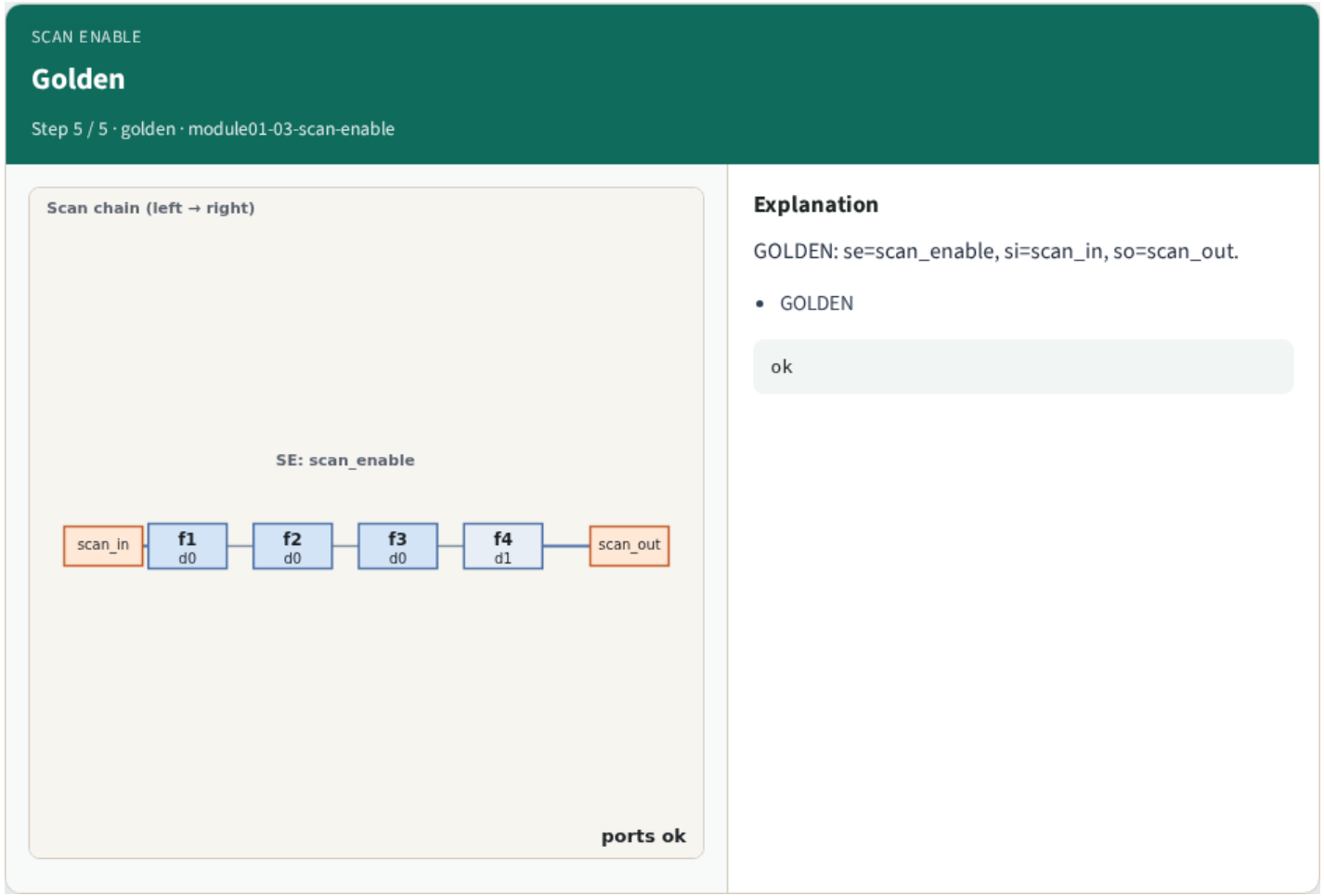
si=scan_in, so=scan_out, se=scan_enable.

- ports

```
3
```

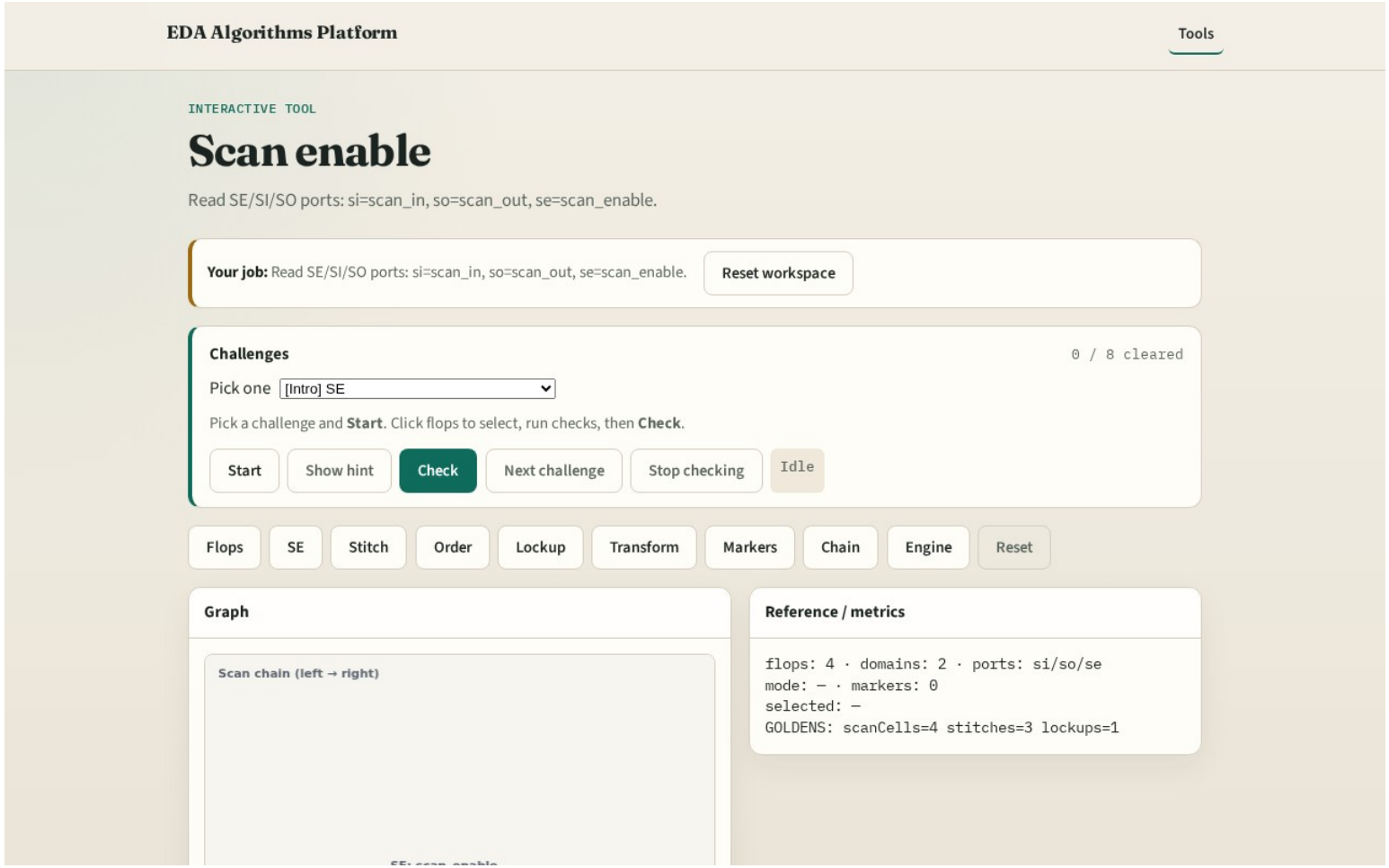
Port names

Golden



Golden

Browser lab track



Browser lab starter

Implement track

- Call scan_enable_port(); assert se/si/so match goldens

Pitfalls

- Confusing SI with functional D
- Wrong port order

Your turn

- Next: chain stitch